Venkata Lakshmi Parimala Pasupuleti

Software Engineer · Partner Integration · Housing & PropTech Domain · REST APIs · AI-Enabled Apps · Python · AWS

+1-703-261-5410 | venkatalakshmiparimala@gmail.com | linkedin.com/in/venkata-lakshmi-parimala-pasupuleti | github.com/venkatalakshmiparimala | Virginia, USA

SUMMARY

Software engineer with 9+ years building the backend systems that platforms like EliseAI integrate with. At Entrata (7+ years, 100K+ users) - a multi-tenant apartment and affordable housing SaaS platform - owned the APIs, data pipelines, event-driven workflows, and compliance systems across the full property management lifecycle: leasing, income eligibility, rent calculation, document workflows, and resident communications. Deep understanding of how housing software works under the hood: data flows, async processing patterns, OAuth-secured integrations, and the business rules that make it complex. Currently at Intuit integrating Anthropic Claude into production financial services - RAG pipelines, structured outputs, AI evaluation, guardrails. Built **Leasel** - an AI platform for lease compliance - because I know the domain well enough to see where AI can help. Strong across REST APIs, webhooks, event-driven architecture, AWS (API Gateway, Lambda, SQS, EventBridge), and cross-functional stakeholder communication. M.S. Data Analytics Engineering, George Mason University (GPA 3.87).

CORE COMPETENCIES

Partner Integration · Solutions Engineering | API Design · REST APIs · Webhooks | System Architecture · Data Flows · Integration Patterns | Event-Driven Architecture · Messaging · Async Processing | AWS (API Gateway · Lambda · SQS · SNS · EventBridge · ECS/EKS) | OAuth2 · OIDC · SAML · JWT · Secure Integrations | Housing & PropTech Domain · Property Management Systems | AI-Enabled Applications · RAG · LLM Integration | Prompt Engineering · Guardrails · AI Evaluation | Python · Java · TypeScript · Node.js | PostgreSQL · Oracle · MongoDB · Redis | Technical Project Management · Cross-Functional Collaboration | Developer Documentation · Playbooks · Tooling | Docker · Kubernetes · GitHub Actions CI/CD | Production Monitoring · Observability · Incident Response

EXPERIENCE

Software Engineer
Jun 2026 - Present

Intuit — Virginia, USA

- Scope and build integrations connecting Anthropic Claude to production Python/FastAPI financial services - data flows from customer records into RAG pipelines, structured-output schemas so downstream services can consume LLM responses reliably, and guardrails with input validation and fallbacks.
- Build LLM evaluation harnesses (LLM-as-judge, golden datasets) to gate releases; coordinate across engineering and product to prioritize gaps in AI integration quality; document patterns and tooling for team reuse. Deploy via Docker and GitHub Actions CI/CD to AWS with CloudWatch monitoring.

Senior Software Engineer
Oct 2020 - Jul 2024

Entrata — Apartment & Affordable Housing SaaS (regulated multi-tenant, 100K+ monthly active users)

- **Built and owned the APIs and data systems that housing platforms integrate with** - leasing workflows, income eligibility calculations, rent processing, HUD compliance pipelines, and document management across thousands of properties and tenant records. Deep understanding of the business rules, data models, and failure modes in housing SaaS backends.
- **Designed and operated event-driven integration architecture** - RabbitMQ-based async message bus for rent updates across thousands of tenant records, with idempotency guarantees, retry logic, and failure routing; AWS S3 integration with PHI encryption (Libsodium, 100% coverage, zero incidents); OAuth-secured API integrations across external data providers.
- Optimized 200+ SQL queries (30% performance gain); Docker microservices redesign (50% scalability improvement); Python/Bash automation eliminating 80% of manual operational work; mentored 5+ engineers; 300+ code reviews across 120+ Agile sprints.

Junior Software Engineer
Sep 2018 - Oct 2020

Entrata India — Pune

- Built property management and compliance modules (Python, PHP, Java, PostgreSQL); delivered large-scale data integration and reconciliation workflows (1.6M records); led AWS S3 cloud migration.

PROJECTS

Leasel - AI Platform for Lease Compliance (Housing Domain)
2026 | Python, LangGraph (8-agent), FastAPI, ChromaDB, CI/CD | github.com/venkatalakshmiparimala/Lease_complaince_rag

- Built from scratch because I know the domain: an 8-agent LangGraph platform that investigates lease compliance questions - OCR ingestion pipeline, hybrid search RAG (TF-IDF + semantic reranking, citation grounding), tool calling with durable state, and guardrails ensuring regulated decisions go to humans not the LLM. CI evaluation harness with LLM-as-judge scoring: 93.6% precision, 1,240 test cases. Three consuming surfaces share one data and retrieval layer.

AgentProbe - LLM Evaluation & Multi-Provider Integration Platform
2025 | Python, FastAPI, PostgreSQL, CI/CD | github.com/venkatalakshmiparimala/AgentProbe

- 15-endpoint FastAPI backend integrating five LLM providers (Anthropic, OpenAI, Azure OpenAI, Groq, Ollama) with cost-aware routing, full tracing, and structured logging per run. Offline golden-dataset evals, LLM-as-judge scoring, and CI regression harness that gates releases on eval thresholds. Documented API contracts and integration patterns for external consumers.

AI on FHIR — Clinical NLP & Healthcare API Integration
2024 | Python, Flask, React, BERT, LLMs, FHIR REST APIs, Docker | github.com/venkatalakshmiparimala/AI_on_FHIR

- Translates English/Spanish natural-language questions into FHIR REST API calls using LLMs, BERT, and embeddings - cut clinical lookup time ~60%. OAuth-secured API integration, HIPAA-style RBAC, containerized with Docker. Integration scope spans patient scheduling, intake forms, and provider communication workflows - exactly the healthcare workflows EliseAI automates.

EDUCATION

M.S. Data Analytics Engineering — George Mason University, Virginia | GPA 3.87 May 2026

B.Tech Computer Science & Engineering — JNTU Anantapur 2017